YAKEEN

NEET 2.0 2027

Physics | Lecture - 03

Units and Measurement

By- Manish Raj (MR Sir)



H/w goal

(1) for Backlog of
Vector → Lecture Uploaded 5hr

(2) Sangha Assignment - 1 (Udm)

(3) Sangh Assignment Vecto - 4

Todays Goal

PhD. on Principle of Homogeneity

U.D.M → Lecture "2"

copy at ghar

MR* Box

P.Q ka value S.I unit me given hai → New/C.G.S unit me value puchha hai ✓

S.I unit me value likho then S.I unit ko C.G.S/new unit me convert karo.

MR* Box

P.Q ka (new unit) ka value S.I unit ke terms me nikalna ho →

P.Q ka new unit likho then new unit ko → S.I me convert karo

Question

H/W

The density of a material in CGS system of units is 4 g cm^{-3}. In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be

[Mains 2011]

1. 0.04
2. 0.4
3. 40
4. 400

$1g' = 100g$

$1cm' = 10cm$

$$d = 4\,\frac{g}{cm^3}$$

$$= 4\,\frac{1}{100} \times \frac{}{\left(\frac{1}{10}\right)^3}$$

Question

In new system of unit, unit of length is 10m, unit of time is 2s, unit of mass is 5 kg, then find unit of torque in new system of unit.

1. 125 Nm
2. 0.125 Nm
3. 8 Nm
4. 8×10^{-3} Nm

Unit of torque $= \frac{1\text{kg}\,(1\text{m})^2}{(1\text{s})^2} = \frac{5\text{kg} \times 25\,\text{m}^2}{4\,\text{s}^2}$

$\frac{1\text{kg}\,(1\text{m})^2}{(1\text{s})^2} = 125 \times \frac{\text{kg m}^2}{\text{s}^2}$

Question

In a new system of units, unit of mass is α kg unit of length is β m and unit of time is γ s. In this system, 10J will be represented as

1. $10\alpha^{-1}\beta^{2}\gamma^{2}$
2. $10\alpha^{-2}\beta^{-1}\gamma^{-2}$
3. $10\alpha^{-1}\beta^{-2}\gamma^{2}$
4. $10\alpha\beta^{2}\gamma^{-2}$

$1\,kg' = \alpha\,kg$

$\text{Energy} = 10\,J$

$= 10 \times \frac{kg\,m^2}{s^2} = 10 \times \frac{1\,kg'}{\alpha}$

Question

Copy

True / false
(a) (b)

(1) A unitless physical quantity may have dimension → false

(2) A dimensionless physical quantity may be unitless → True

Ex - Angle, coeffici of fric

(3) A physical quantity have unit must have dimension → false (B.)

Ex - Angle

(4) A physical quantity have dimension may have unit → false

(5) A P.Q. have dimensio must have unit → True

Mpt Box Copy at ghcy

Dimension (2nd Floor)	(A) Ans then must be dimensionless	(B) Ans may or may not have dimensin.	(C) A P.Q does not have dimension then ??	(D) A P.Q have dimension then ??
Unit (1st Floor)	(C) A P.Q. does not hav unit hence unitless then ??	(C) A P.Q. have Unit then ??	(C) Ans may or may not have unit.	(D) Ans must have Unit

Question

COPY

If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time? **[JEE Main 2022]**

1. RC (T)
2. L/R (T)
3. $\sqrt{LC}$ (T)
4. L/C

$\omega = 2\pi f = \frac{1}{T}$



$R = \omega L = \frac{1}{\omega C} \Longrightarrow (\omega L = \frac{1}{\omega C})$

$R = \omega L$

$\frac{R}{L} = \frac{1}{T}$

$T = \frac{L}{R}$

$R = \frac{1}{\omega C}$

$RC = T$

$LC = \frac{1}{\omega^2}$

$LC = T^2$

$T = \sqrt{LC}$

find value of given exp^{n}.

$4kg + 8m = ??$ ✗ ⟶ ulta think karo

$A + B \Rightarrow A = B$ dimensionally

$4kg + 8kg = 12kg$ ✓

$A - B + C \Rightarrow A = B = C$ dimensionally

$8hr - 3kg =$ ✗

$4kg > 2m$ ✗

$\left[\frac{3kg}{5m} = \frac{3}{5} kg/m = \text{linear mass density} \right.$

$\frac{5m}{2\,sec} = \frac{5}{2} m/s$ ✓ (speed)

$5kg \times 2sec = 10 kg\,sec$ ✓ $\left. \right]$

Principle of Homogenity → We can add, subtract or compair two P.Q. only when they have same dimension.

We can Multiply or divide two P.Q which have same or different dimension

MR* Box

equation me +ve, -ve or >, < ka sigh hai sab ko hta ke equal kar do only for dimensional analysis.

MR* Boy :-

equation me +, −, >, <
hai to sab ko hta ke equal
kar do (dimensionally)

MR* BOX! –

Torque and Work can not add Physically but they can add dimensionally.

$\tau = \vec{r} \times \vec{F}$ $\quad W = \vec{F} \cdot \vec{r}$

① Force $F = \alpha t + \beta x - \gamma t^2$ find dimension of α, β & γ where
$x \rightarrow$ Position $t =$ time

MR^*

$F = \alpha t = \beta x = \gamma t^2$

$F = \alpha t$ $\quad F = \beta x$ $\quad F = \gamma t^2$

$F = \alpha t$ $\quad \beta = \frac{F}{x}$ $\quad \gamma = \frac{F}{t^2}$

$\alpha = \frac{F}{t} = \frac{MLT^{-2}}{T}$ $\quad = \frac{MLT^{-2}}{L}$ $\quad = MLT^{-4}$

$= MLT^{-3}$ $\quad = MT^{-2}$

Question

Copy

Electric field in a certain region is given by $\vec{E} = \left(\frac{A}{x^2}\hat{i} + \frac{B}{y^3}\hat{j}\right)$. The SI unit of A and B are:

[JEE Main 2023]

$F = qE$

$E = \frac{F}{q} = N/C$

1. $Nm^3 C^{-1}$; $Nm^2 C^{-1}$

Electric field $\vec{E} = \frac{A}{x^2}\hat{i} = \frac{B}{y^3}$

2. ✓ $Nm^2 C^{-1}$; $Nm^3 C^{-1}$

$A = Ex^2$

$= Nm^2/C$

3. $Nm^3 C$; $Nm^2 C$

4. $Nm^2 C$; $Nm^3 C$

Copy

(3) Velocity $V = \frac{At}{(B-x)}$; find dimension of A & B.

$B = x$

$B = M^0 L^1 T^0$

$V = \frac{At}{x}$

$A = \frac{Vx}{t} = \frac{LT^{-1} \times L}{T}$

$A = L^2 T^{-2}$ ✓

$4kg - 3m = X$

$4kg - 1kg = 3kg$ ✓

Question

(4)

The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a, b and c are constants. The dimensions of a, b and c are **(2006)**

1. [L], [LT] and [LT^{-2}]
2. [LT^{-2}], [L] and [T]
3. [L^2], [T] and [LT^{-2}]
4. [LT^{-2}], [LT] and [L]

$v = at = \frac{b}{t}$

$a = \frac{v}{T} = LT^{-2}$

$t = c$

$c = T^1$

⑤ $\left(P + \frac{b}{V^2}\right) = a$ find dimension of a & b

P → Pressure

V → volume

$P = \frac{b}{V^2} = a$

$a = P$ (Press $= \frac{F}{A} = ML^{-1}T^{-2}$ ✓

$b = PV^2$

$= ML^{-1}T^{-2}(L^3)^2$

$= ML^5T^{-2}$

Question

6

An equation is given here $\left(P + \frac{a}{v^2}\right) = b\frac{\theta}{V}$ where P = Pressure, V = Volume and θ = Absolute temperature. If a and b are constants, the dimensions of a will be

(1996)

1. $[ML^{-5}T^{-1}]$
2. $[ML^{5}T^{1}]$
3. $[ML^{5}T^{-2}]$
4. $[M^{-1}L^{5}T^{2}]$

$$P = \frac{a}{V^2} = \frac{b\theta}{V}$$

$$a = PV^2$$

$$= mL^5T^{-2}$$

$$b = \frac{PV}{\theta}$$

$$= \frac{mL^{-1}T^{-2}(L^3)}{K}$$

Question

Not Copy

Assertion: Work = Torque is dimensionally correct but not physically. (True)

Reason: Dimensional correctness of an equation ensures its physical correctness. X

1. If both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
2. If both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
3. ✓ If Assertion is True but the Reason is False.
4. If both Assertion & Reason are false.

Some dimension less function

① Angle (θ)

$$\text{dimension of Angle} = M^0 L^0 T^0 = 1 \times 1 \times 1 = 1$$

* $P = \frac{F}{A} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$

↳ Physics mei ese 11th detail class nahi → MR Physics → Question Bank Book

$$\text{Arc length} = R\theta$$

$$\text{Arc length} = R \times 1 = L^1$$

② Trigonometric function :-

$$Y = \sin\theta = \frac{P}{H} = M^0 L^0 T^0 = 1$$

⑦ If P(momentum) $= \alpha \sin(\beta)$ then find dimension of α & β

$\sin\beta \to 1$

Sol^n $P = \alpha \times 1$ $\quad \beta = 1$

$\alpha = MLT^{-1}$ $\quad \beta = M^0 L^0 T^0 = 1$

(3) exponetial function

$$y = e^{x} = (2.71)^{x}$$

→ dimensionless hote!

→ dimension less hota hai

Question (8) If Velocity $V = \alpha e^{(\beta t)}$ then find dim of α & β.

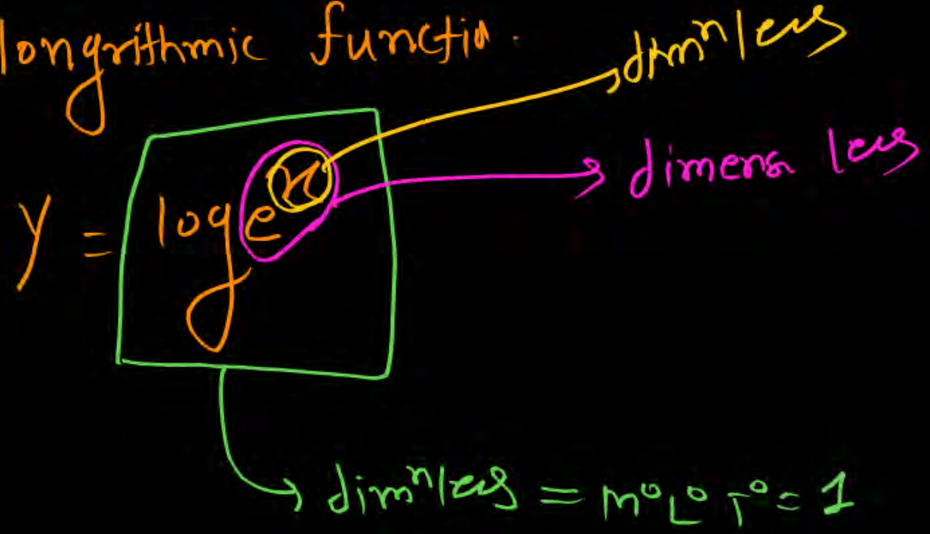
Sol^n

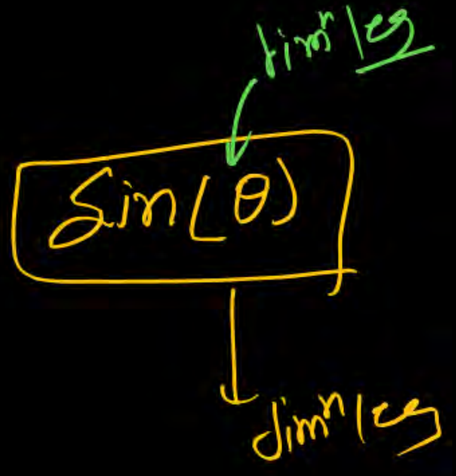
$$V = \alpha \qquad \beta \cdot t = 1$$

$$\alpha = LT^{-1} \qquad \beta = \frac{1}{t} = T^{-1}$$

✓ =

(4) logarithmic function.





copy

MR* Box :-

Thara Bhai MR, θ, $\sin\theta$, e^x
$\log e^x$ sab ko 1 likh dega.

Question

Q9 The time dependence of a physical quantity p is given by $p = p_0 \exp(-at^2)$, where a is a constant and t is the time. The constant a **(1993)**

1. is dimensionless
2. has dimensions $[T^{-2}]$
3. has dimensions $[T^2]$
4. has dimensions of p

$P = P_0 e^{-(at^2)}$

"$at^2 = 1$"

$a = \frac{1}{t^2} = T^{-2}$

Question

Assertion: The unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ have units of distance and dimensions $[M^0 L^1 T^0]$

Reason: The product of a scalar and a vector is a new scalar.

1. If both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
2. If both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
3. If Assertion is True but the Reason is False.
4. If both Assertion & Reason are false.

Question

Assertion: Force is not added with pressure.

Reason: Quantities having different dimension are not added or subtracted.

1. If both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
2. If both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
3. If Assertion is True but the Reason is False.
4. If both Assertion & Reason are false.

Question

Copy

$Y = A \sin(\omega t - kx + \phi)$, find dimension of A, ω, k and ϕ where y and x is position.

Angle

$Y = A$ ✓

$\omega t = kx = \phi = 1$

$\omega t = 1$

$\omega = \frac{1}{T} = T^{-1}$

$\phi = 1 (M^0 L^0 T^0)$

$kx = 1$

$k = \frac{1}{x} = L^{-1}$

Question

Copy

(NEET) (IIT)

Velocity $V = \frac{\alpha}{\beta} e^{-\alpha t}$ then find dimension of α and β.

$V = \frac{\alpha}{\beta}$

$\alpha t = 1$

$\alpha = T^{-1}$

$\beta = \frac{\alpha}{V}$

$= \frac{T^{-1}}{LT^{-1}} = L^{-1}$

Question

No Cor

If velocity $v = \alpha t + \beta x - \gamma xt$, then find dimension of α, β and γ.

$V = \alpha t = \beta x = \gamma xt$

Question

Velocity $V = \frac{\alpha}{\beta} e^{-\alpha t}$ then find dimension of α and β.

Question

Momentum of object is given as $P = \alpha t[1 - \beta^{\gamma x}]$ then, find dimension of α, β and γ. Where t is time x is position.

$1 = \beta^{\gamma x}$

$\Rightarrow \gamma x = 1$

$\gamma = \frac{1}{x} = L^{-1}$

$\hookrightarrow \beta = 1$

$\beta = M^0 L^0 T^0$

$\Rightarrow P = \alpha t$

$\alpha = \frac{P}{t}$

$= \frac{MLT^{-1}}{T}$

$= MLT^{-2}$

Question

If $Y = A \sin\left[\frac{2\pi}{\lambda}(ct - x)\right]$. Find dimension of c.

$ct = x$

$c = \frac{x}{t} = LT^{-1}$ ✓

Question

Momentum $P = \sin(\alpha t^3 + \beta x^2)$ find dimension of (i) $\alpha \cdot \beta$ (ii) α/β

Sol:

$$\alpha t^3 = \beta x^2 = 1$$

$$\alpha = \frac{1}{t^3}$$

$$\beta = \frac{1}{x^2}$$

Question

Copy

Velocity V = A sin $(x\sqrt{\alpha t})$, find dimension of α.

Soln

A = V

Angle

$x\sqrt{\alpha t} = 1$

Square both side

$x^2 \alpha t = 1^2$

$\alpha = \frac{1}{t x^2} = L^{-2} T^{-1}$

Question

Equation of force $F = \frac{a - x}{bt^2}$ then find dimension of a and b.

$F = \frac{x}{bt^2} \Rightarrow a = x$

$b = \frac{x}{Ft^2}$

dimension of differential term :-

Change in length $\Delta l = l_f - l_i = M^0 L^1 T^0$

Small change in length $dl = M^0 L^1 T^0$

$$\rightarrow \frac{dy}{dx} = \left[\frac{y}{x}\right]$$

$$\leftrightarrow \frac{d^2y}{dx^2} = \frac{y}{x^2} = \frac{y}{x^2} \text{ (dimension)}$$

$$\rightarrow \frac{d^3y}{dx^3} = \frac{y}{x^3}$$

(0) If y = force & x = velocity then find dimension of

$$\frac{dy}{dx} = \frac{y}{x} = \frac{MLT^{-2}}{LT^{-1}} = MT^{-1}$$

$$\frac{d^2y}{dx^2} = \frac{y}{x^2} = \frac{MLT^{-2}}{(LT^{-1})^2} = ML^{-1}$$ ✓

$$\int y\,dx = y \times x = F \times V = MLT^{-2}(LT^{-1})$$

$$= ML^2T^{-3}$$

Question

$$\int \frac{dx}{\sqrt{2ax - x^2}} = b$$; find dimension of a and b if x is distance.

Question

Assertion: If x and y are the distances along x and y axes respectively then the dimensions of $\frac{d^3y}{dx^3}$ is $M^0\,L^{-2}\,T^0$.

Reason: Dimensions of $\int_a^b y\,dx$ is $M^0L^2T^0$.

1. If both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
2. If both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
3. If Assertion is True but the Reason is False.
4. If both Assertion & Reason are false.

Question Copy

Force acting on object is proportional to square of velocity then find dimension of proportional constant.

Soln

$$F \propto v^2$$

$$F = bv^2$$

$$b = \frac{F}{v^2} = \frac{mLT^{-2}}{(LT^{-1})^2} = \frac{mLT^{-2}}{L^2T^{-2}}$$

$$= mL^{-1}$$

Physically	Ans(A) Physically must be wrong.	Ans (B) may or may not correct	(C) An eqn is physically wrong.	(D) An equation is physically correct then ??
dimensionally	(A) An equation is dimensionally wrong then ??	(B) An equation is dimensionally correct then ??	Ans → may or may not correct	Ans → dimensionally must be correct :- ✓

By Force = Pressure + Velocity

Work = Torque + energy

$S = ut$ ✓

MR SCAM (+1)

$S_{nth} = u(1) + \frac{a}{2}(2n-1)$ (1 sec)

dimension Correct

→ disp^{m} in n^{th} sec.

Formula formation by dimension analysis :—

Application of dimensional analysis

① Gf. Time Period of Simple Pendulum depend on length of pendulum (l). & acc^n due to gravity (g) then derive relation (formula) of Time Period in terms of "l" & "g".

Soln

$$T \propto l^a$$

$$T \propto g^b$$

$$T = l^a g^b \quad \rightarrow ①$$

$$M^0 L^0 T^1 = L^a (LT^{-2})^b$$

$$M^0 L^0 T^1 = L^{a+b} T^{-2b}$$

Compair Powr of 'L' & Time

$$0 = a+b$$

$$a = -b$$

$$a = -\left[-\tfrac{1}{2}\right]$$

$$\boxed{a = 1/2}$$

$$1 = -2b$$

$$\boxed{b = -\frac{1}{2}}$$

Put a & b

$$T = l^{1/2} g^{-1/2} = \sqrt{\frac{l}{g}}$$ ✓

Application of dimensional analysis

① gf Time Period of Simple Pendulum depend on length of pendulum (l) & acc^n due to gravity (g) then derive relation formula of Time Period in terms of l & g.

② gf we consider length (l) and acc^n due to gravity (g) are fundamental P.Q. then find dimension of Time interms of l & g.

③ gf in new System of unit of length 10m & unit of acc^n is $5 m/s^2$ then find unit of time.

all 3- are diffrent Question but Same Solution ✓

Question

$v = L^1T^{-1}M^0$ $\quad F = MLT^{-2}$

$a = M^0L^1T^{-2}$

Force acting on object moving on circular path depends on mass (m), radius (r) and velocity V then derive force in terms of them.

Sol:

$$F \propto m^a r^b v^c$$

$$MLT^{-2} = M^a (L)^b (LT^{-1})^c$$

$$MLT^{-2} = M^a L^{b+c} T^{-c}$$

$a = 1$ $\quad +2 = +c$ $\quad C = 2$

$b + c = 1$

$b + 2 = 1$

$b = 1 - 2 = -1$

$$F = m^1 r^{-1} v^2$$

$$\left\{ F = \frac{mv^2}{r} \right\}$$

Q. If g Consider radius, mass & velocity as a fundamental P.Q. then find dimension Force in terms of them

$$F = m^a r^b v^c$$

Question

If force (F), acceleration (a) and time t is used as a fundamental P.Q. then find dimension of length in terms of them:

1. $F^0 a^1 T^2$
2. $F a^2 T^2$ ✗
3. $F a^2 T^0$ ✗
4. $F^0 a T$ ✗

$LT^{-2} \times T = LT^{-1}$

$l = F^{\alpha} a^{\beta} t^{\gamma}$

$(MLT^{-2})^{\alpha} \cdot (LT^2)^{\beta} (T)^{\gamma} = LT^{-2} (T^2)^2$

Rough medium

$s = \frac{1}{2} a t^2$

$l = aT^2$

Question

If force, acceleration and time are basic fundamental P.Q. then find dimension of energy.

1. $F^2 A^{-1} T$
2. $F A T^2$
3. $F A T^{-2}$
4. $F A^{-1} T$

Question

If energy E, velocity V and time T are taken as fundamental units, the dimensional formula for surface tension is

1. $[E\ V^{-2}\ T^{-2}]$
2. $[E^{-2}\ V\ T^{-2}]$
3. $[E^{-2}\ V\ T^{-2}]$
4. $[E^{-2}\ V^{-2}\ T^{2}]$

Question

If maximum acceleration of oscillating particle is $\alpha$and maximum velocity is β, then find time period [NEET 2013]

1. $2\pi\frac{\alpha}{\beta}$
2. $2\pi\alpha\beta$
3. $2\pi\frac{\beta}{\alpha}$
4. $\frac{2\pi\alpha^2}{\beta^2}$

Question

On the basis of dimension, decide which of the following relation for displacement of a particle is not correct [NCERT]

1. $y = a \sin\left(\frac{2\pi t}{T}\right)$
2. $y = a \sin\left(\frac{vt}{\lambda}\right)$
3. $y = \sqrt{2}a \sin\left(\frac{2\pi t}{T}\right) \cos\left(\frac{2\pi x}{\lambda}\right)$
4. $y = \frac{x}{t} \sin(\lambda t)$

Question

If force (F), acceleration (a) and time t is used as a fundamental P.Q. then find dimension of length in terms of them:

1. $F^0 a^1 T^2$
2. $F a^2 T^2$
3. $F a^2 T^0$
4. $F^0 a T$

Question

If energy E, velocity V and time T are taken as fundamental units, the dimensional formula for surface tension is

1. $[E\ V^{-2}\ T^{-2}]$
2. $[E^{-2}\ V\ T^{-2}]$
3. $[E^{-2}\ V\ T^{-2}]$
4. $[E^{-2}\ V^{-2}\ T^{2}]$

Question

If force F, area A and density D are taken as the fundamental units, the representation of Young's modulus 'Y' will be:

1. $[F^{-1} A^{-1} D^{-1}]$
2. $[F A^{-2} D^{2}]$
3. $[F A^{-1} D]$
4. $[F A^{-1} D^{0}]$

Question

Plank's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length? **[NEET 2016]**

1. $\frac{\sqrt{hG}}{c^{3/2}}$
2. $\frac{\sqrt{hG}}{c^{5/2}}$
3. $\sqrt{\frac{hc}{G}}$
4. $\frac{\sqrt{Gc}}{h^{3/2}}$

Question

A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and θ is charge]

1. $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

2. $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

3. $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$

4. $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$

Try all H/W at HomE

Thank

You